Sindh Science Learning Portal

Class 5 · Science

Chapter 4

Environmental Pollution



Learn what pollution is, how air, water and land get polluted, and how we can cut waste and keep our surroundings clean.

Following the Sindh Textbook Board curriculum

sindh-science-portal.vercel.app

CHAPTER 4

Environmental Pollution

What we will learn Learn what pollution is, how air, water and land get polluted, and how we can cut waste and keep our surroundings clean.

1 What is pollution?

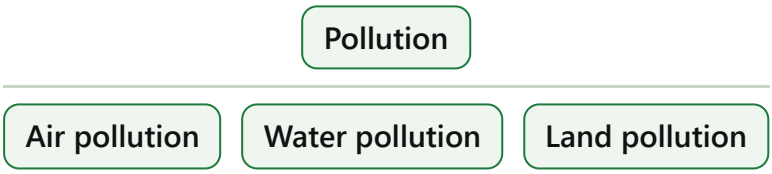
When things that are **harmful** to people, animals and plants get mixed into air, water or land, we call it **pollution**.

The harmful things that cause pollution are called **pollutants**, such as smoke, dirty water, plastic bags and chemicals from factories.

Most pollution is caused by the things **people** do.

There are three main kinds of pollution: **air pollution**, **water pollution** and **land pollution**.

Diagram: the parts and what they do



Pollution (آلودگي · آلودگی)
Harmful things getting mixed into air, water or land.

Air pollution (هوا جي آلودگي · ہوائی آلودگی)
Smoke from vehicles, factories and burning rubbish. It causes coughing and asthma.

Water pollution (پاڻيءَ جي آلودگي · آبی آلودگی)
Sewage, chemicals and rubbish in water. It causes cholera and typhoid.

Land pollution (زمين جي آلودگي · زمینی آلودگی)
Throwing plastic, bottles and rubbish on the ground. It spoils soil and underground water.

2 Air pollution

Causes of air pollution:

- **Smoke** from cars, buses and rickshaws.
- Smoke and gases from **factory** chimneys.
- **Burning rubbish** and burning fuels such as wood, coal and oil.

Breathing polluted air causes **coughing**, **asthma**, burning eyes, headaches and lung diseases. In big cities like Karachi, smoke from vehicles makes air pollution very high.



Picture: Adina Voicu · CC0 · Wikimedia Commons

3 Water and land pollution

Causes of **water pollution**: letting dirty water from homes (**sewage**) into rivers or the sea without cleaning it, chemicals from factories, and pesticides and fertilisers washing off farms.

Drinking polluted water causes **cholera**, **typhoid** and diarrhoea. Fish and other living things also die in polluted water.

Land pollution happens when rubbish such as plastic bags, bottles, paper and tin cans is thrown on the ground.

Heaps of rubbish spoil the soil and the water under the ground. Animals get sick from eating plastic, and flies and mosquitoes breed in rubbish.

4 Biodegradable and non-biodegradable waste

- **Biodegradable waste**: rots away into the soil after some time because of microbes. Examples: fruit and vegetable peels, leftover food, dry leaves and paper.
- **Non-biodegradable waste**: does not rot even in a hundred years. Examples: **plastic**, glass, tin cans and polythene.

Plastic that reaches rivers and the sea harms fish, turtles and birds.

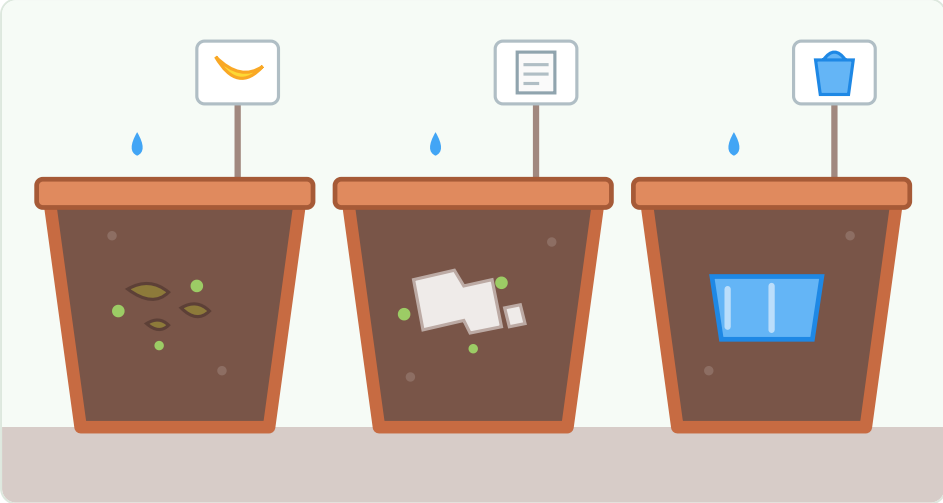
We can reduce pollution: **reduce**, **reuse** and **recycle**. Use cloth bags, do not burn rubbish, put rubbish in the bin, walk or share rides, and **plant trees**.

Key words

English	سنڌي	اردو
pollution	آلودگي	آلودگی
pollutant	آلوده ڪندڙ	آلودہ کار
environment	ماحول	ماحول

English	سنڌي	اردو
smoke	دونهون	دھواں
sewage	سيوريج	سیوریج
waste	ڪچرو	کچرا
biodegradable	ڳرندڙ ڪچرو	گلنے والا کچرا
non-biodegradable	نه ڳرندڙ ڪچرو	نہ گلنے والا کچرا
recycle	ري سائيڪل	ری سائیکل
asthma	دمو	دمہ

Experiment: Which waste rots away in soil?



You need

- Three small pots filled with soil
- A banana or vegetable peel
- A piece of paper
- A piece of plastic bag
- Water and labels

Steps

1. Bury one item about 5 centimetres deep in each pot.
2. Label each pot with the name of the item.
3. Sprinkle a little water every other day to keep the soil damp.
4. After three weeks, dig carefully and look at all three items.
5. Write down which item rotted and which stayed the same.

Safety: Wash your hands with soap after touching soil and waste. Do not use glass or sharp things.

What did we learn? The peel rotted a lot and the paper became soft and started to break up, but the plastic stayed exactly the same. Microbes in the soil break down biodegradable waste but cannot break down plastic. So we should use less plastic.

Practice questions

Circle the correct answer

1. **What is pollution?**

   (a) Harmful things getting into air, water or land
   (b) Planting trees
   (c) Rain falling
   (d) Putting rubbish in a bin

2. **What are the harmful things that cause pollution called?**

   (a) Manure
   (b) Pollutants
   (c) Biodegradables
   (d) Vaccines

3. **Which of these causes air pollution?**

   (a) Riding a bicycle
   (b) Planting trees
   (c) Burning rubbish
   (d) Walking

4. **Which illness can breathing polluted air cause?**

   (a) Asthma
   (b) Tooth decay
   (c) A broken bone
   (d) Athlete's foot

5. **What is dirty water from homes called?**

   (a) Rainwater
   (b) Sewage
   (c) Steam
   (d) Ice

6. **Which disease can come from drinking polluted water?**

   (a) Cholera
   (b) Weak eyesight
   (c) Asthma
   (d) None

7. **Which of these is biodegradable?**

   (a) Plastic bottle
   (b) Glass
   (c) Banana peel
   (d) Tin can

8. **What should we use instead of a plastic bag?**

   (a) Another plastic bag
   (b) A cloth bag
   (c) Polythene
   (d) Nothing, just burn it

9. **What does recycling mean?**

   (a) Making new things from old things
   (b) Burning rubbish
   (c) Throwing rubbish in a river
   (d) Buying more plastic

Write true or false

10. **Most pollution is caused by people.** _____

11. **Burning rubbish makes the air clean.** _____

12. **Pesticides washing off farms can pollute water.** _____

13. **Plastic rots into the soil in a few days.** _____

14. **Planting trees helps keep the air clean.** _____

Match the columns

15. **Match each cause with the kind of pollution.**

1. Smoke from cars		(a) Water pollution
2. Sewage in the sea		(b) Land pollution
3. Heaps of rubbish		(c) Air pollution

Match the columns

16. **Match each piece of waste with its kind.**

1. Dry leaves		(a) Non-biodegradable
2. Plastic bag		(b) Biodegradable

Worksheet: Environmental Pollution

Name: _____ Date: _____

A. Fill in the blanks

Word box: Plastic · asthma · pollutants · recycling · sewage

1. The harmful things that cause pollution are called _____ .
2. Dirty water from homes is called _____ .
3. Polluted air can cause _____ .
4. _____ does not rot even in a hundred years.
5. Making new things from old things is called _____ .

B. Put ✓ for true and ✗ for false

1. ☐ Smoke from factories pollutes the air.
2. ☐ Fish live happily in polluted water.
3. ☐ Fruit peels are biodegradable.
4. ☐ Burning rubbish is a good way to reduce pollution.

C. Match column A with column B (write the number in the brackets)

Column A	Column B
1. Air pollution	( ) Plastic bags
2. Water pollution	( ) Vegetable peels
3. Land pollution	( ) Smoke from vehicles
4. Biodegradable waste	( ) Factory chemicals

D. Answer in short

1. Name the three kinds of pollution.

_____

_____

2. What is the difference between biodegradable and non-biodegradable waste?

_____

_____

3. Write three ways to reduce pollution.

_____

_____

E. Draw and label

Make a poster to keep your school clean. Draw a rubbish bin on it and write a slogan.

FOR TEACHERS

Answer key

Chapter 4: Environmental Pollution

Practice questions

1. (a) Harmful things getting into air, water or land
2. (b) Pollutants
3. (c) Burning rubbish
4. (a) Asthma
5. (b) Sewage
6. (a) Cholera
7. (c) Banana peel
8. (b) A cloth bag
9. (a) Making new things from old things
10. True
11. False
12. True
13. False
14. True
15. 1 → (c), 2 → (a), 3 → (b)
16. 1 → (b), 2 → (a)

Worksheet

A. Fill in the blanks: 1. pollutants · 2. sewage · 3. asthma · 4. Plastic · 5. recycling

B. Put ✓ for true and ✗ for false: 1. ✓ · 2. ✗ · 3. ✓ · 4. ✗

C. Match column A with column B (write the number in the brackets): (3) Plastic bags · (4) Vegetable peels · (1) Smoke from vehicles · (2) Factory chemicals

D. Answer in short:

1. Air pollution, water pollution and land pollution.
2. Biodegradable waste rots into the soil; non-biodegradable waste does not rot for many years.
3. Use cloth bags, do not burn rubbish and plant trees.

Picture credits

- Chapter 4: Adina Voicu · CC0 · commons.wikimedia.org/wiki/File:Smoke_Pollution.jpg

This book was made on the Sindh Science Learning Portal. Free and without ads. sindh-science-portal.vercel.app